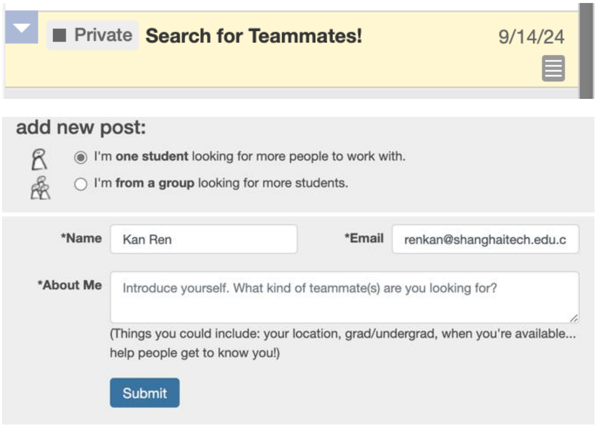
Project announcement

Group

- You are free to establish the team with up to 5 team members
  - You may use the Piazza "search for teammates" function

- Schedule with due time
  - April 30th: form groups
    - Submit your group info in the following online sheet: https://docs.qq.com/sheet/DZUtDZmVOcmRLUWRz?tab=BB08J2
  - May 6th: proposal submission due
  - June 5th (Thursday, Week 16)
    - Course time: final presentation
    - 11:59pm: report submission due



Project announcement

Topic

- Each group needs to select one topic from academic journals or conferences (CCF-A or B https://www.ccf.org.cn/Academic_Evaluation/By_category).
- The requirements of the topic are:
  - The paper you choose should be **relevant** to the course content, but the algorithm examples in the textbook and slides cannot be regarded as your topic.
  - It should solve a problem in a specific domain.

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Topic

- Bad examples:
  - A Max-Flow/Min-Cut implementation.
  - An AI algorithm with PyTorch implemented gradient descent optimization.
- Good examples:
  - Paper "Network-Flow-Based Efficient Vehicle Dispatch for City-Scale Ride Hailing Systems" use network flow algorithms to tackle the problem in large-scale dispatching system.
  - Paper "Progressive Divide-and-Conquer via Subsampling Decomposition for Accelerated MRI" use divide conquer to tackle the problem in MRI reconstruction.
  - Paper "MurTree: Optimal Decision Trees via Dynamic Programming and Search" use the dynamic programming to optimize the classification tree.

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Submission requirements

- 1. The original paper.
- 2. Program source code, focusing on the algorithm part of the paper.
- 3. An executable program or demo video.
- 4. The experimental report should include but not be limited to:
  - Analysis of the correctness of the algorithm.
  - Formal description of the algorithm (pseudo code).
  - Complexity analysis of the algorithm.
  - Program running environment and running results.
  - Consideration and improvement of the algorithm in the paper.
  - The division of labor in your team.

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Grading

- 10% of the total grade.

Criteria

- Relevance to this course.
- Soundness, substance.
- Quality of the report and presentation.